Minimax regret for conditionally similar tests

Andrés Aradillas Fernández[a]*

[a]Department of Economics, Massachusetts Institute of Technology
77 Massachusetts Avenue, Building E52-300
Cambridge, MA 02139, United States

Abstract

We construct approximately minimax-regret tests in a scalar Gaussian instrumental-variables experiment. Existing computational approaches maximize power against a fixed alternative distribution, including a mixture representing weighted average power, while optimizing over null distributions to enforce size. Conditional similarity enforces size within each best response, allowing us to optimize weights over alternatives to minimize worst-case power regret. Stochastic mirror descent and averaging yield finite-iteration guarantees with exact conditional similarity. We establish bounds for exact and simulated powers, control discretization error in total variation, and illustrate the procedure numerically.

Keywords: Conditional similarity; minimax regret; weak instruments; stochastic mirror descent.
JEL classification: C12, C14.

1 Introduction

Conditional similarity controls rejection probabilities in the presence of nuisance parameters but generally leaves many tests to choose from. We select among these tests by minimizing their largest power loss relative to the conditionally similar power envelope. This criterion is useful when instruments may be weak and no single test maximizes power at every alternative.

Our contribution is to use conditional size control to optimize power regret across alternatives. Elliott et al. (2015), henceforth EMW, fix a weighting distribution over alternatives; weighted average power then equals power against a single mixture distribution. Their numerical search concerns a least-favorable distribution under the null. Aradillas Fernández et al. (2025) develop stochastic mirror descent for the corresponding finite-null, single-alternative problem. This formulation also accommodates a fixed alternative mixture representing a weighted-average-power criterion.

We instead impose conditional similarity within every best response. This removes the need to optimize null weights to enforce size and allows the iterative search to focus on alternatives at which power falls furthest below its conditionally similar envelope. Weighted average power remains the best-response objective, but its weights are updated to solve a minimax-regret problem rather than specified in advance as the final power criterion. Our result concerns the conditionally similar class in the Gaussian experiment; it does not claim to dominate EMW's procedures over the unrestricted class of level-α tests.

For a finite alternative grid, we combine conditional Neyman–Pearson best responses with exponential weight updates. Averaging preserves exact conditional similarity and yields explicit regret bounds for exact and simulated power calculations. Andrews (2016) studies minimax regret within a class of conditional linear combination tests, while Montiel Olea (2020) characterizes admissible similar tests. We optimize over the full conditionally similar class in the scalar experiment considered here.

*Corresponding author. Email: aradilla@mit.edu.

2 The testing problem

We observe a pair of scalar reduced-form estimates satisfying

$$\begin{pmatrix}\hat{\delta}\\ \hat{\pi}\end{pmatrix} \sim N\left(\begin{pmatrix}\pi\beta\\ \pi\end{pmatrix}, \Sigma\right), \qquad \Sigma \succ 0, \tag{1}$$

where Σ is known and does not depend on (β, π). We test $H_0 : \beta = \beta_0$ at level $\alpha \in (0, 1)$, with unrestricted nuisance parameter $\pi \in \mathbb{R}$. Define

$$S = \hat{\delta} - \beta_0\hat{\pi}, \qquad a = \mathrm{Cov}(\hat{\pi}, S)/\mathrm{Var}(S), \qquad U = \hat{\pi} - aS. \tag{2}$$

The transformation is invertible. Let $\sigma_S^2 = \mathrm{Var}(S)$ and $\sigma_U^2 = \mathrm{Var}(U)$. At $\theta = (\beta, \pi)$, the variables S and U are independent normals with means $m(\theta) = \pi(\beta - \beta_0)$ and $\tau(\theta) = \pi - am(\theta)$ and variances σ_S^2 and σ_U^2. Under the null, $S \mid U = u \sim N(0, \sigma_S^2)$, independently of π.

A randomized test is a measurable rejection probability $\varphi(s, u) \in [0, 1]$. Writing f_0 for the $N(0, \sigma_S^2)$ density, we consider

$$C_\alpha = \left\{\varphi : \int \varphi(s, u) f_0(s)\, ds = \alpha \text{ for Lebesgue-almost every } u\right\}. \tag{3}$$

Because the null density of U is everywhere positive, this restriction is equivalent to conditional similarity for every π. It implies unconditional rejection probability α.

For alternatives $\theta \in \Theta \subseteq \{(\beta, \pi) : \beta \neq \beta_0\}$, define the power envelope, regret, and minimax value by

$$b(\theta) = \sup_{\varphi \in C_\alpha} \mathbb{E}_\theta\varphi, \qquad \mathcal{R}(\varphi, \theta) = b(\theta) - \mathbb{E}_\theta\varphi, \qquad V_\Theta = \inf_{\varphi \in C_\alpha} \sup_{\theta \in \Theta} \mathcal{R}(\varphi, \theta). \tag{4}$$

We first take $\Theta = \Theta_I = \{\theta_1, \ldots, \theta_I\}$, with $I \geq 2$, and write $V_I = V_{\Theta_I}$ and $b_i = b(\theta_i)$.

3 Conditional best responses and weight updates

For weights w in the probability simplex Δ^{I-1}, let $g_w = \sum_i w_i g_i$, where g_i is the joint density of (S, U) at θ_i. The best response maximizes weighted average power:

$$\varphi_w \in \arg\max_{\varphi \in C_\alpha} \sum_{i=1}^{I} w_i \mathbb{E}_{\theta_i}\varphi. \tag{5}$$

The weights w are optimization variables for worst-case regret. Thus each best response uses a single alternative mixture, but the mixture changes across iterations as the algorithm reallocates weight toward high-regret alternatives. The constraint in (3) separates across u. Conditional Neyman–Pearson optimization therefore rejects for large values of

$$L_w(s, u) = \frac{g_w(s, u)}{f_0(s)} = \sum_{i=1}^{I} w_i f_U(u; \tau_i) \exp\left(\frac{m_i s}{\sigma_S^2} - \frac{m_i^2}{2\sigma_S^2}\right), \tag{6}$$

where $m_i = m(\theta_i)$, $\tau_i = \tau(\theta_i)$, and $f_U(\cdot; \tau_i)$ is the $N(\tau_i, \sigma_U^2)$ density. Specifically, $\varphi_w = \mathbf{1}\{L_w > c_w(u)\} + \gamma_w(u)\mathbf{1}\{L_w = c_w(u)\}$, with threshold and boundary randomization chosen to satisfy (3). For a single alternative, the optimal rule is one-sided in S, giving

$$b_i = \Phi\left(\Phi^{-1}(\alpha) + |m_i|/\sigma_S\right), \tag{7}$$

Algorithm 1 Conditionally similar testing by stochastic mirror descent

Require: Grid Θ_I, size α, iterations K, draws N, step size η.

Compute envelopes b_i using (7); set $w_{1,i} = 1/I$.
for $k = 1, \dots, K$ **do**
    Construct φ_k from (6), imposing conditional size α.
    Draw fresh $Y_{k,i,n} \sim g_i$ independently for all i, n.
    Set $\widehat{p}_{k,i} = N^{-1} \sum_n \varphi_k(Y_{k,i,n})$ for every i.
    Set $w_{k+1,i} \propto w_{k,i} \exp\{\eta(b_i - \widehat{p}_{k,i})\}$; normalize.
end for
return $\bar{\varphi}_K = K^{-1} \sum_k \varphi_k$.

where Φ is the standard normal distribution function. When $m_i = 0$, the envelope equals α.

The function $F(w) = \sup_{\varphi \in C_\alpha} \sum_i w_i(\mathbb{E}_{\theta_i}\varphi - b_i)$ is convex, with subgradient $G_i(w) = \mathbb{E}_{\theta_i}\varphi_w - b_i$. Negative-entropy mirror descent thus increases weights in proportion to estimated regret. Starting from $w_{1,i} = 1/I$, at iteration k we compute $\varphi_k = \varphi_{w_k}$ and set

$$\widehat{p}_{k,i} = \frac{1}{N} \sum_{n=1}^{N} \varphi_k(Y_{k,i,n}), \qquad \widehat{G}_{k,i} = \widehat{p}_{k,i} - b_i, \qquad w_{k+1,i} = \frac{w_{k,i} \exp(-\eta \widehat{G}_{k,i})}{\sum_j w_{k,j} \exp(-\eta \widehat{G}_{k,j})}. \tag{8}$$

Conditional on the preceding iterations, the draws $Y_{k,i,n} \sim g_i$ are independent across i and n and fresh at each iteration. Hence $\widehat{G}_k$ is conditionally unbiased. Exact-power updates replace $\widehat{p}_{k,i}$ by $\mathbb{E}_{\theta_i}\varphi_k$.

We return $\bar{\varphi}_K = K^{-1} \sum_{k=1}^{K} \varphi_k$, implemented by selecting an iteration uniformly and applying its test. Simulation draws and this random selection are independent of the data being tested. Conditional on the simulation history, $\bar{\varphi}_K \in C_\alpha$. Thus simulation affects regret but does not affect conditional size when best responses are computed exactly.

4 Finite-iteration guarantees

Proposition 1. *For the exact conditional best responses in* (5)*, the following statements hold for every* $K \geq 1$ *and* $\eta > 0$*.*

1. *With exact-power updates,*
$$\max_i \mathcal{R}(\bar{\varphi}_K, \theta_i) \leq V_I + \frac{\log I}{\eta K} + \frac{\eta}{8}. \tag{9}$$

2. *With the simulated updates in* (8)*, for any* $\delta \in (0, 1)$*, with probability at least* $1 - \delta$ *over the simulation draws,*
$$\max_i \mathcal{R}(\bar{\varphi}_K, \theta_i) \leq V_I + \frac{\log I}{\eta K} + \frac{\eta}{2} + \sqrt{\frac{2\log((I+1)/\delta)}{NK}}. \tag{10}$$

Proof. Write $r_{k,i} = \mathcal{R}(\varphi_k, \theta_i) \in [0, 1]$. Best-response optimality gives $w_k^\top r_k = \inf_{\varphi \in C_\alpha} \sum_i w_{k,i} \mathcal{R}(\varphi, \theta_i) \leq V_I$. The exponential-weights inequality for rewards in $[0, 1]$ yields

$$\max_i \frac{1}{K} \sum_k r_{k,i} \leq \frac{1}{K} \sum_k w_k^\top r_k + \frac{\log I}{\eta K} + \frac{\eta}{8}. \tag{11}$$

Linearity of power proves (9).

For simulated updates, $\widehat{G}_{k,i} \in [-1,1]$. The same inequality, now with range length two, gives, simultaneously for all i,

$$\frac{1}{K}\sum_k (r_{k,i} - w_k^\top r_k) \le \frac{\log I}{\eta K} + \frac{\eta}{2} + \frac{1}{K}\sum_k \Delta_{k,i} - \frac{1}{K}\sum_k w_k^\top \Delta_k, \tag{12}$$

where $\Delta_{k,i} = \widehat{G}_{k,i} - G_i(w_k)$. Conditional Hoeffding bounds and iteration over the simulation history imply $\Pr(K^{-1}\sum_k \Delta_{k,i} > x) \le \exp(-2NKx^2)$. The same bound holds for $-K^{-1}\sum_k w_k^\top \Delta_k$: conditional independence gives squared summand ranges totaling $\sum_i w_{k,i}^2/N \le 1/N$ at each iteration. A union bound over these $I+1$ deviations, with $x = \sqrt{\log((I+1)/\delta)/(2NK)}$, bounds the two noise terms in (12) by $2x$. Using $w_k^\top r_k \le V_I$ and linearity proves (10). □

With exact powers, $\eta = \sqrt{8\log I/K}$ gives error $\sqrt{\log I/(2K)}$. With simulated powers, an error tolerance $\epsilon > 0$ and failure probability δ are attained by

$$K = \left\lceil \epsilon^{-2}\left(\sqrt{2\log I} + \sqrt{\frac{2\log((I+1)/\delta)}{N}}\right)^2 \right\rceil, \qquad \eta = \sqrt{\frac{2\log I}{K}}. \tag{13}$$

The guarantees concern the average test, not the best response to average weights. They require no minimax interchange: weighted regret is bounded by V_I directly. Indeed, (11) also establishes $V_I = -\min_{w\in\Delta^{I-1}} F(w)$ by letting K increase, combining weak duality with the bound on average weighted regret; continuity of F ensures attainment.

5 Discretization and implementation

Corollary 1. *Suppose $\Theta_I \subseteq \Theta$ and $\sup_{\theta\in\Theta}\min_i d_{\mathrm{TV}}(P_\theta, P_{\theta_i}) \le \rho$, where $d_{\mathrm{TV}}(P,Q) = \sup_A |P(A) - Q(A)|$. Any $\varphi \in C_\alpha$ with grid regret at most $V_I + \epsilon$ satisfies*

$$\sup_{\theta\in\Theta} \mathcal{R}(\varphi,\theta) \le V_\Theta + \epsilon + 2\rho. \tag{14}$$

Proof. For every $[0,1]$-valued test, $|\mathbb{E}_\theta\varphi - \mathbb{E}_{\theta'}\varphi| \le d_{\mathrm{TV}}(P_\theta, P_{\theta'})$. Taking suprema gives the same bound for $|b(\theta) - b(\theta')|$. Regret therefore changes by at most twice total variation. Approximation by the grid and $V_I \le V_\Theta$ give (14). □

In the Gaussian experiment, compact alternative sets admit arbitrarily fine total-variation grids by continuity of the mean. The unrestricted alternative space requires a separate tail argument; a finite grid alone does not establish uniform optimality there.

Implementation requires conditional critical values and evaluations of (6) at simulated observations. Equation (13) separates iteration error from simulation error, but is not a bound on total computing time. Numerical errors in critical values or best-response optimization are outside Proposition 1 and must be assessed separately. Subject to exact conditional best responses, the procedure delivers a conditionally similar test with a prescribed finite-grid regret guarantee.

6 A numerical illustration

We set $\beta_0 = 0$, $\Sigma = I_2$, and $\alpha = 0.05$, so that $S = \hat{\delta}$ and $U = \hat{\pi}$. Our grid contains $\theta_1 = (-3, 0.5)$ and $\theta_2 = (1,2)$, giving $(m_1, m_2) = (-1.5, 2)$ and $(\tau_1, \tau_2) = (0.5, 2)$. The corresponding power envelopes are

Table 1: Power and regret on the two-alternative grid

Test	Size	Power at θ_1	Power at θ_2	Max. regret
Equal-tail test	0.050	0.323	0.516	0.123
Initial best response	0.050	0.361	0.567	0.082
SMD average, 100 iterations	0.050	0.362	0.566	0.081
SMD average, 1000 iterations	0.050	0.364	0.563	0.078

Notes: The equal-tail test rejects when $|S| > \Phi^{-1}(0.975)$. The initial best response uses $w = (1/2, 1/2)$. SMD rows average all tests through the indicated iteration of the same run. Powers are evaluated by numerical integration; size follows from the tail allocation. Maximum regret is relative to the pointwise envelopes in (7).

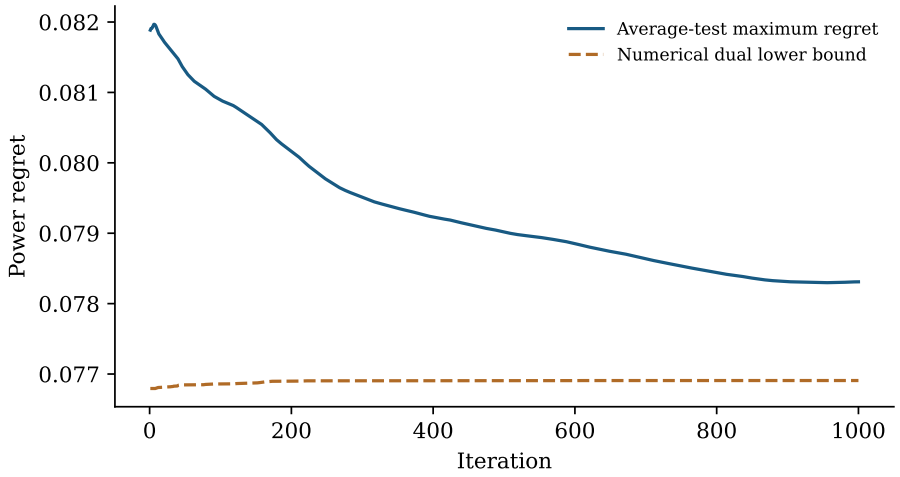


Figure 1: Regret along the SMD iterations. We report the maximum regret of the average test and the running maximum of $w_k^\top r_k$, evaluated by numerical integration. With exact best responses and integration, the latter bounds V_I from below. OpenAI Codex assisted with the plotting code; the figure is generated reproducibly from computed results.

0.4424 and 0.6388. This deliberately small grid isolates the trade-off between power against negative and positive effects.

For these two alternatives, the likelihood ratio in (6) is strictly convex in s. The best response rejects in two tails. If $q_w(u)$ is its left-tail null probability, the critical values satisfy

$$\ell_w(u) = \Phi^{-1}(q_w(u)), \qquad h_w(u) = \Phi^{-1}(1 - \alpha + q_w(u)), \qquad L_w(\ell_w(u), u) = L_w(h_w(u), u). \tag{15}$$

We solve the last equality by bisection over $q \in [10^{-14}, \alpha - 10^{-14}]$. The resulting rule has conditional size $q + (\alpha - q) = \alpha$ by construction. Larger values of u favor the second alternative and shift rejection probability toward the right tail.

We execute Algorithm 1 for $K = 1{,}000$ iterations with $N = 512$ independent draws per alternative per iteration, $\eta = \sqrt{2\log(2)/K}$, and random seed 20261102. We evaluate power separately by integrating the conditional rejection probability with 320-node Gauss–Hermite quadrature. Doubling the quadrature order from 160 to 320 changes the computed iteration-specific powers by less than 10^{-12}. These are results from one simulation run, rather than averages across repeated training runs. OpenAI Codex assisted in designing and implementing this numerical illustration, running the computations, and drafting their interpretation; the author reviewed and verified the code and results. The accompanying source records the implemented procedure and all simulation settings.

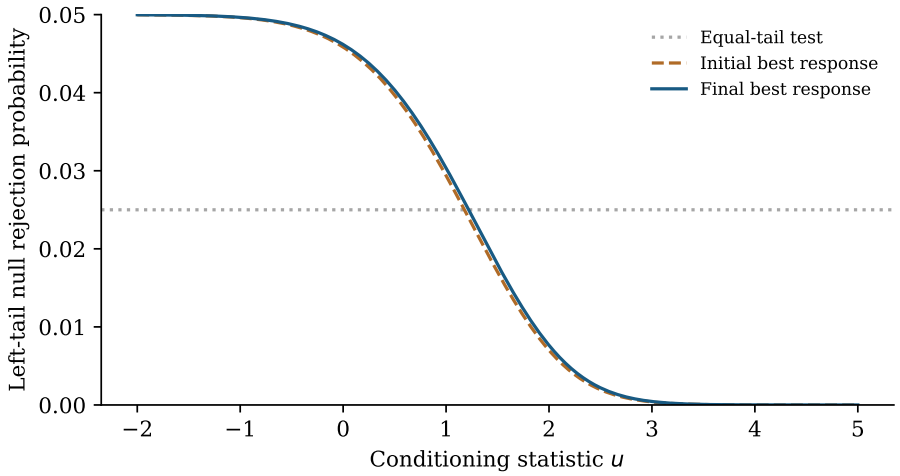


Figure 2: Conditional tail allocation. Left-tail null rejection probabilities for the initial and final individual best responses are compared with the equal-tail test. The returned test averages all iterates. OpenAI Codex assisted with the plotting code; the figure is generated reproducibly from computed results.

Table 1 shows that conditioning improves power at both alternatives relative to the equal-tail test. Figure 2 shows how the conditioning statistic shifts rejection probability between tails. Subsequent weight updates increase power at θ_1 while sacrificing some power at θ_2, reducing the largest regret. After 1,000 iterations, maximum regret is 0.0783, compared with 0.0819 initially and 0.1228 for the equal-tail test. Figure 1 reports a numerical dual lower bound of 0.0769, leaving a numerical gap of 0.0014. This diagnostic concerns the chosen grid and numerical best responses; it is not a uniform bound over unexamined alternatives. The replication package reproduces both figures, the table, and their underlying numerical results (Aradillas Fernández, 2026).

7 Conclusion

Conditional similarity allows us to shift the numerical search from null distributions enforcing size to alternative distributions determining worst-case power regret. Relative to EMW's fixed weighted-average-power criterion, the alternative weights are chosen endogenously to balance regret within the conditionally similar class. Conditional Neyman–Pearson best responses and stochastic mirror descent deliver finite-grid guarantees while preserving exact conditional size. Extensions beyond the grid require control of discretization error, and numerical implementations require accurate best responses.

Data and code availability

We use simulated data and no external or confidential data. The Python implementation, dependency versions, random seed, computed results, and reproduction instructions are available in the public GitHub replication repository (Aradillas Fernández, 2026): `https://github.com/aaf101/SMD-for-Conditional-Similarity`. The version tagged `v1.0.0` records the replication package for this manuscript. The README describes how to reproduce the figures and table and compile the paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI Codex to substantially rewrite and organize an existing manuscript; assist with mathematical exposition, proof arguments, and literature comparisons; develop, execute, and check the Python numerical illustration; generate its figures and table from computed results; and prepare and check the LaTeX source and compiled manuscript. The numerical figures were produced by plotting program output, rather than by generative image synthesis. The author directed this work and subsequently reviewed and verified the AI-assisted text, mathematical arguments, references, code, and numerical results, and edited the content as needed. The author takes full responsibility for the content of the article.

References

Andrews, I. (2016). Conditional linear combination tests for weakly identified models. *Econometrica*, 84(6):2155–2182. https://doi.org/10.3982/ECTA12407.

Aradillas Fernández, A. (2026). Replication materials for Minimax regret for conditionally similar tests. GitHub, version 1.0.0 [software and simulation results]. https://github.com/aaf101/SMD-for-Conditional-Similarity/tree/v1.0.0.

Aradillas Fernández, A., Blanchet, J., Montiel Olea, J. L., Qiu, C., Stoye, J., and Tan, L. (2025). Approximate least-favorable distributions and nearly optimal tests via stochastic mirror descent. *arXiv preprint arXiv:2511.16925*. https://doi.org/10.48550/arXiv.2511.16925.

Elliott, G., Müller, U. K., and Watson, M. W. (2015). Nearly optimal tests when a nuisance parameter is present under the null hypothesis. *Econometrica*, 83(2):771–811. https://doi.org/10.3982/ECTA10535.

Montiel Olea, J. L. (2020). Admissible, similar tests: A characterization. *Econometric Theory*, 36(2):347–366. https://doi.org/10.1017/S0266466619000215.